

## Project Update (Real-time object tracking)

### **Week 1 (Sep 23 – Sep 30)**

We selected our project topic after forming project group (7). Our topic is Find 10 videos online showing objects moving from one place to another. Then, create a machine learning model that can accurately track the locations of these moving things in real-time.

### **Week 2 (Oct 01 – Oct 07)**

I did not do anything this week.

### **Week 3 (Oct 08 – Oct 14)**

This week, along with my two partners, started exploring what are the to-dos of our project. The first to-do is that we need to understand what way / road to choose. Option A is classical CV (open computer vision), option B is deep learning (HRNet, YOLO V2). We also have the term 'real-time location' in our project. Whether "real-time" means GPS coordinates or whether it means using an algorithm to track object's position frame by frame while our video runs is what we will discuss with our instructor in the upcoming weeks.

### **Week 04 (Oct 15 – Oct 21)**

We looked at the functions i.e., models that would best fit our project. My instructor said to try using ML at first. And if we cannot do it, then opt for DL. So, I looked into different Machine Learning pipelines for tracking during this week. Mainly KNN method for detecting moving objects by stationary camera. This method is called Background subtraction. Another algorithm is Lucas-Kanade for sparse feature tracing. I learnt about all these methods of object detection this week.

### **Week 05 (Oct 22 – Oct 28)**

The earlier week we looked into various tracking ML pipelines. This week the target was to look for public videos that I can use. Primarily, the videos have a very important criterion. It should have reasonable or sufficient spatial details. By having so, it will separate the objects in them from their background. Also, another important factor is that there is variety in the movement patterns of the different video-objects. We drafted many videos out of which we will finalize

our 10 videos (all videos have variety, spatial details, 720 p resolution). Also, I learnt how to use SIFT and ORB as well to detect the features. If we decide to shift to SIFT/ORB approach, then we could do so in the future. I also learnt the Kalman filter Kalman filter and the Hungarian algorithm to address multi-object tracking problems. The model needs to first understand what the object is. When we give the model any new video to evaluate, it needs to first identify if it is a dog, a cat, a car or something else. Then it will start tracking it's coordinates frame by frame.

## **Week 06 (Oct 29 – Nov 04)**

We finalized the full system pipeline.